YUYAY ANALYTICS

Jay Antonio Oliver, MSc

DATA SCIENCE & GEOSPATIAL ANALYTICS LEAD

Machine learning · Spatial intelligence · Environmental data engineering

United States / Ecuador | jayjayoliver@hotmail.com | English · Spanish

PROFILE

Data scientist and geospatial specialist with 6+ years of experience building scalable analytical pipelines, predictive models, and spatial decision-support solutions across environmental, energy, and research domains. Combines Python, SQL, GIS, remote sensing, and reproducible machine learning to translate complex datasets into actionable intelligence.

TECHNICAL EXPERTISE

Machine learning — XGBoost, Random Forest, classification and regression, feature engineering, cross-validation, SHAP, scikit-learn.

Data engineering — Python, SQL, pandas, NumPy, APIs, ETL, automation, relational data models, Linux and Git.

Geospatial analytics — GeoPandas, Rasterio, Shapely, GIS, remote sensing, spatial overlays and land-use analysis.

PROFESSIONAL EXPERIENCE

Leidos | Geospatial Data Scientist | 2024–2026 — Developed Python and SQL pipelines and predictive models for energy and environmental datasets. Analyzed over 7 million well records to identify high-confidence orphaned-well candidates; supported Appalachian Basin modeling and coal ash mineral prediction.

Baylor University | Lab Manager / Geospatial Data Scientist | 2021–2024 — Led Python-based analytical and research workflows; developed reproducible data pipelines, geospatial models and automated highway imagery collection. Contributed primary Python development to a peer-reviewed land-use data publication.

USGS | Spatial Analyst | 2021 — Processed and visualized more than two decades of environmental data for land-cover change assessment.

WWF | Data Scientist / Statistical Analyst | 2020 — Integrated spatial, biodiversity and socioeconomic data for deforestation and land-use analysis across multi-country datasets.

NASA DEVELOP | Remote Sensing Data Scientist | 2019–2020 — Automated ingestion and processing of more than 900 Earth observation datasets for regional environmental analysis in Peru and Central America.

SELECTED ANALYTICAL APPLICATIONS

Environmental risk and prioritization — Large-scale classification of orphaned oil and gas wells using spatial and operational predictors.

Land-use and ecosystem dynamics — Harmonization and analysis of multi-source environmental and geospatial data.

Remote sensing and monitoring — Satellite time-series processing and geospatial data integration for environmental decision support.

Automated analytics — Reproducible ETL and data quality workflows for complex scientific and operational sources.

EDUCATION

MSc Environmental Economics & Environmental Management | University of York | 2018

BSc Environmental Science & Policy | University of South Florida | 2014